R123 | Unique VAL-selected checkpoint | VAL and TEST

Task	VAL Top1 %	TEST Top1 %	VAL cost	TEST cost	VAL regret %	TEST regret %
TSP	46.91	46.62	8.0251	7.9715	0.421	0.421
CVRP	47.24	47.62	18.158	18.011	0.691	0.678
ATSP	77.07	77.04	1.6386	1.6352	0.370	0.351
OVRP	40.89	40.73	7.4623	7.4467	0.641	0.633
VRPB	81.00	80.53	9.355	9.3506	0.396	0.382
VRPL	37.24	37.64	12.432	12.349	0.396	0.399
VRPTW	79.89	81.02	18.819	18.689	0.298	0.291
OVRPTW	83.84	83.93	10.143	10.093	0.259	0.236
OVRPB	88.51	88.11	6.3573	6.371	0.265	0.304
OVRPL	41.47	41.93	7.4376	7.4635	0.655	0.623
VRPBL	78.71	79.56	9.3688	9.3962	0.436	0.432
VRPBTW	78.51	78.89	19.947	19.664	0.466	0.438
VRPLTW	78.84	79.33	18.765	18.801	0.320	0.317
OVRPBL	89.24	89.27	6.3695	6.3756	0.273	0.268
OVRPBTW	83.60	84.89	10.66	10.659	0.322	0.294
OVRPLTW	83.53	82.91	10.157	10.086	0.273	0.267
VRPBLTW	79.36	79.73	19.795	19.706	0.438	0.450
OVRPBLTW	83.22	83.62	10.61	10.662	0.345	0.322
ALL18	71.06	71.30	11.417	11.374	0.404	0.395

One locked full18 checkpoint; no TEST/LIB policy or checkpoint selection.
TRAIN here = source-record-certified metaTRAIN10886/task, not full TRAIN81000; historical holdouts development-exposed.
SBS = fixed R84 original TRAIN81000 choice. ALL18 = equal task weight; raw cost macro retained only for routing history.
Native error and relative-cost regret are different budgets. An exact-cost tie may be a native error with zero regret.

R123 | TRAIN | all required native metrics

Task	N	Top1 %	Mean cost	SBS cost	Oracle cost	vs SBS %	vs Oracle %	Actual regret %
TSP	10,886	56.21	10.331	10.376	10.287	-0.43564	0.43275	0.43842
CVRP	10,886	47.593	18.486	18.605	18.392	-0.63977	0.51136	0.6746
ATSP	10,886	76.18	1.6401	1.6789	1.6341	-2.309	0.37096	0.37035
OVRP	10,886	40.998	7.4456	7.4542	7.3987	-0.11588	0.63351	0.64656
VRPB	10,886	80.241	9.355	9.354	9.3172	0.01092	0.40574	0.4089
VRPL	10,886	37.369	12.417	12.425	12.369	-0.062256	0.39068	0.39787
VRPTW	10,886	80.323	18.802	18.802	18.748	-0.0022383	0.28726	0.29836
OVRPTW	10,886	83.41	10.115	10.115	10.089	-0.0012463	0.26308	0.26269
OVRPB	10,886	88.637	6.3634	6.3633	6.3474	0.0015555	0.25262	0.25982
OVRPL	10,886	40.63	7.4467	7.455	7.4001	-0.112	0.6292	0.64136
VRPBL	10,886	80.029	9.4062	9.4052	9.3659	0.01119	0.43082	0.42984
VRPBTW	10,886	79.157	19.685	19.687	19.595	-0.010242	0.45836	0.44282
VRPLTW	10,886	80.351	18.735	18.736	18.678	-0.0046798	0.30525	0.32062
OVRPBL	10,886	88.444	6.3569	6.3568	6.3397	0.0018379	0.27101	0.28094
OVRPBTW	10,886	83.226	10.615	10.616	10.577	-0.0027101	0.35948	0.33337
OVRPLTW	10,886	83.511	10.12	10.12	10.093	-0.0028359	0.26275	0.26028
VRPBLTW	10,886	79.313	19.766	19.767	19.675	-0.0042309	0.4599	0.44287
OVRPBLTW	10,886	83.281	10.611	10.611	10.572	-0.0037281	0.36297	0.33641
ALL18	195,948	71.606	11.539	11.552	11.493	-0.2045	0.39376	0.40256

One locked full18 checkpoint; no TEST/LIB policy or checkpoint selection.
TRAIN here = source-record-certified metaTRAIN10886/task, not full TRAIN81000; historical holdouts development-exposed.
SBS = fixed R84 original TRAIN81000 choice. ALL18 = equal task weight; raw cost macro retained only for routing history.
Native error and relative-cost regret are different budgets. An exact-cost tie may be a native error with zero regret.

R123 | VAL | all required native metrics

Task	N	Top1 %	Mean cost	SBS cost	Oracle cost	vs SBS %	vs Oracle %	Actual regret %
TSP	4,500	46.911	8.0251	8.0454	7.9905	-0.25218	0.43216	0.42141
CVRP	4,500	47.244	18.158	18.269	18.064	-0.60733	0.52223	0.69089
ATSP	4,500	77.067	1.6386	1.6769	1.6325	-2.287	0.37265	0.37023
OVRP	4,500	40.889	7.4623	7.4696	7.4158	-0.096858	0.62795	0.64068
VRPB	4,500	81	9.355	9.3549	9.3187	0.0013411	0.38902	0.39585
VRPL	4,500	37.244	12.432	12.438	12.383	-0.051305	0.3887	0.39599
VRPTW	4,500	79.889	18.819	18.819	18.766	-0.0020972	0.28188	0.29837
OVRPTW	4,500	83.844	10.143	10.144	10.117	-0.0038148	0.26178	0.25899
OVRPB	4,500	88.511	6.3573	6.3573	6.3411	0	0.2551	0.26501
OVRPL	4,500	41.467	7.4376	7.4457	7.3904	-0.10927	0.63809	0.65531
VRPBL	4,500	78.711	9.3688	9.3715	9.3284	-0.028686	0.43382	0.43584
VRPBTW	4,500	78.511	19.947	19.95	19.851	-0.01568	0.4847	0.46569
VRPLTW	4,500	78.844	18.765	18.766	18.708	-0.005136	0.30503	0.31959
OVRPBL	4,500	89.244	6.3695	6.3701	6.3528	-0.0089964	0.26282	0.273
OVRPBTW	4,500	83.6	10.66	10.661	10.623	-0.0073713	0.34893	0.32199
OVRPLTW	4,500	83.533	10.157	10.157	10.13	-0.0030756	0.26943	0.27296
VRPBLTW	4,500	79.356	19.795	19.797	19.706	-0.0099044	0.44992	0.43799
OVRPBLTW	4,500	83.222	10.61	10.61	10.571	0.0014869	0.36714	0.34459
ALL18	81,000	71.06	11.417	11.428	11.372	-0.19366	0.39396	0.40358

One locked full18 checkpoint; no TEST/LIB policy or checkpoint selection.
TRAIN here = source-record-certified metaTRAIN10886/task, not full TRAIN81000; historical holdouts development-exposed.
SBS = fixed R84 original TRAIN81000 choice. ALL18 = equal task weight; raw cost macro retained only for routing history.
Native error and relative-cost regret are different budgets. An exact-cost tie may be a native error with zero regret.

R123 | TEST | all required native metrics

Task	N	Top1 %	Mean cost	SBS cost	Oracle cost	vs SBS %	vs Oracle %	Actual regret %
TSP	4,500	46.622	7.9715	7.9896	7.9376	-0.22646	0.42752	0.42054
CVRP	4,500	47.622	18.011	18.125	17.919	-0.62695	0.51812	0.67808
ATSP	4,500	77.044	1.6352	1.6742	1.6294	-2.3317	0.3516	0.35099
OVRP	4,500	40.733	7.4467	7.4557	7.4005	-0.12203	0.62379	0.63257
VRPB	4,500	80.533	9.3506	9.3518	9.3153	-0.012545	0.37936	0.38217
VRPL	4,500	37.644	12.349	12.356	12.301	-0.056304	0.39202	0.39917
VRPTW	4,500	81.022	18.689	18.691	18.637	-0.0056271	0.27947	0.29134
OVRPTW	4,500	83.933	10.093	10.093	10.068	-0.0031503	0.24443	0.2357
OVRPB	4,500	88.111	6.371	6.3708	6.3522	0.0033738	0.2962	0.30389
OVRPL	4,500	41.933	7.4635	7.4725	7.418	-0.12056	0.61336	0.62343
VRPBL	4,500	79.556	9.3962	9.3964	9.356	-0.002307	0.42962	0.43189
VRPBTW	4,500	78.889	19.664	19.666	19.578	-0.0075049	0.44226	0.43817
VRPLTW	4,500	79.333	18.801	18.803	18.745	-0.0093338	0.30001	0.31729
OVRPBL	4,500	89.267	6.3756	6.3757	6.3593	-0.0014387	0.25693	0.26835
OVRPBTW	4,500	84.889	10.659	10.66	10.625	-0.0059344	0.32199	0.29448
OVRPLTW	4,500	82.911	10.086	10.086	10.059	-0.0033872	0.26612	0.26744
VRPBLTW	4,500	79.733	19.706	19.707	19.613	-0.0074228	0.47071	0.44988
OVRPBLTW	4,500	83.622	10.662	10.662	10.626	0.00023798	0.34446	0.3223
ALL18	81,000	71.3	11.374	11.385	11.33	-0.19661	0.38655	0.39487

One locked full18 checkpoint; no TEST/LIB policy or checkpoint selection.
TRAIN here = source-record-certified metaTRAIN10886/task, not full TRAIN81000; historical holdouts development-exposed.
SBS = fixed R84 original TRAIN81000 choice. ALL18 = equal task weight; raw cost macro retained only for routing history.
Native error and relative-cost regret are different budgets. An exact-cost tie may be a native error with zero regret.

R123 | Original native LIB | all required metrics, same locked checkpoint

Dataset	N	Top1 %	Mean cost	SBS cost	Oracle cost	vs SBS %	vs Oracle %	Actual regret %
TSPLIB	49	18.367	8.2184	8.2404	8.1422	-0.26757	0.9361	0.94636
CVRPLIB	100	56	67.43	67.245	66.945	0.27403	0.72379	0.76164

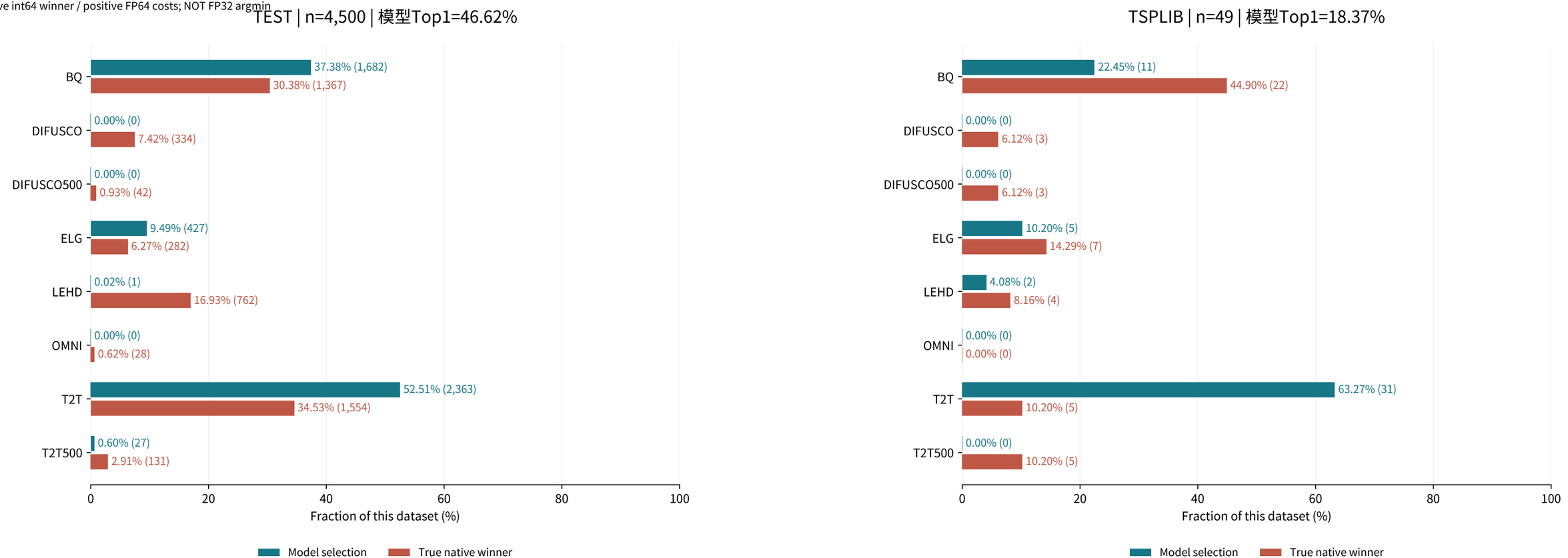
One locked full18 checkpoint; no TEST/LIB policy or checkpoint selection.
TRAIN here = source-record-certified metaTRAIN10886/task, not full TRAIN81000; historical holdouts development-exposed.
SBS = fixed R84 original TRAIN81000 choice. ALL18 = equal task weight; raw cost macro retained only for routing history.
Native error and relative-cost regret are different budgets. An exact-cost tie may be a native error with zero regret.

R123 | TSP | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

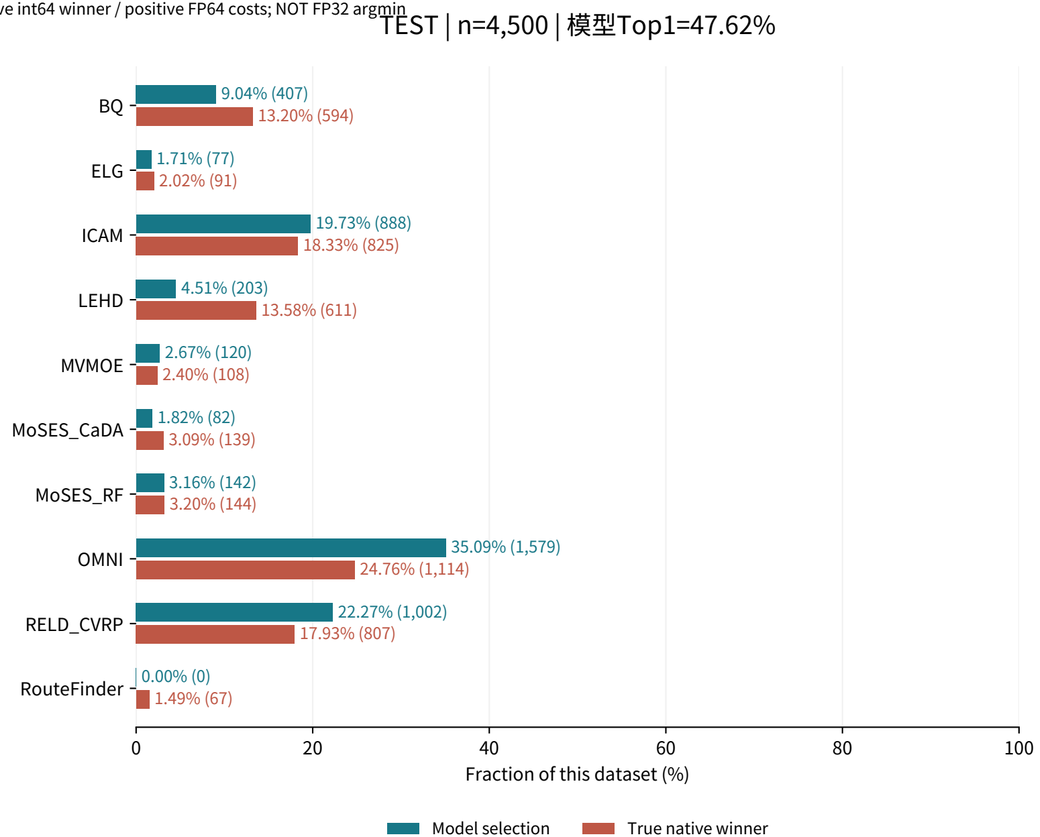
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

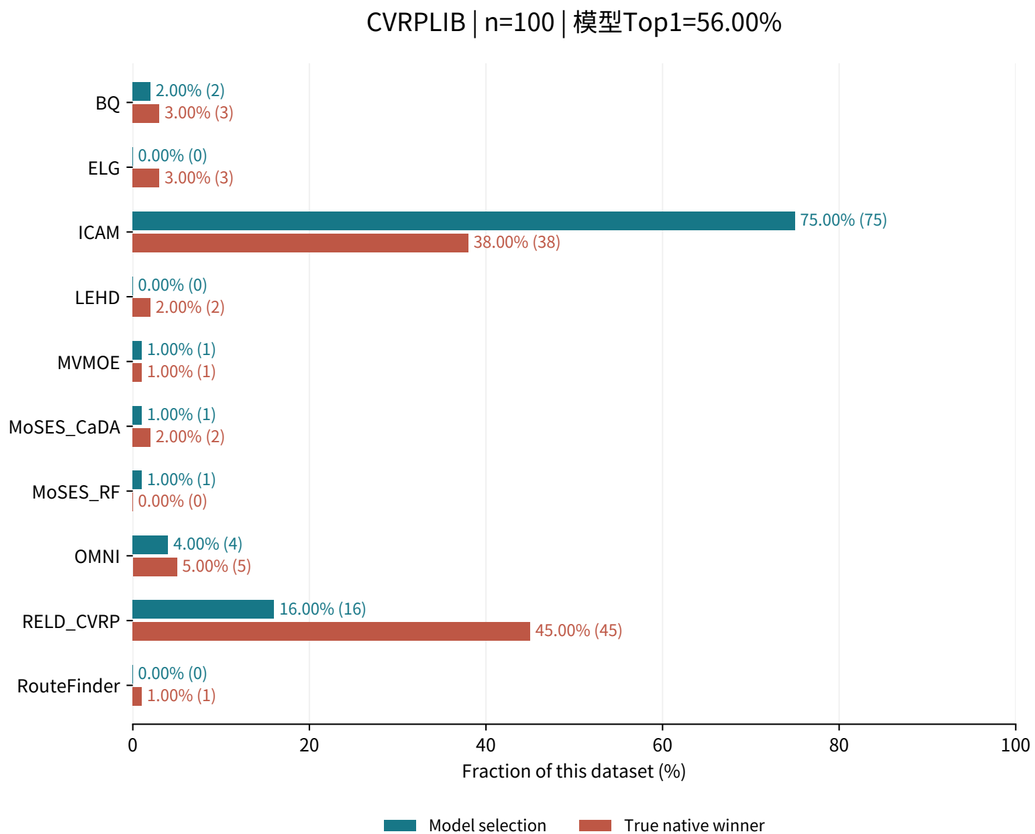
R123 | CVRP | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin





Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

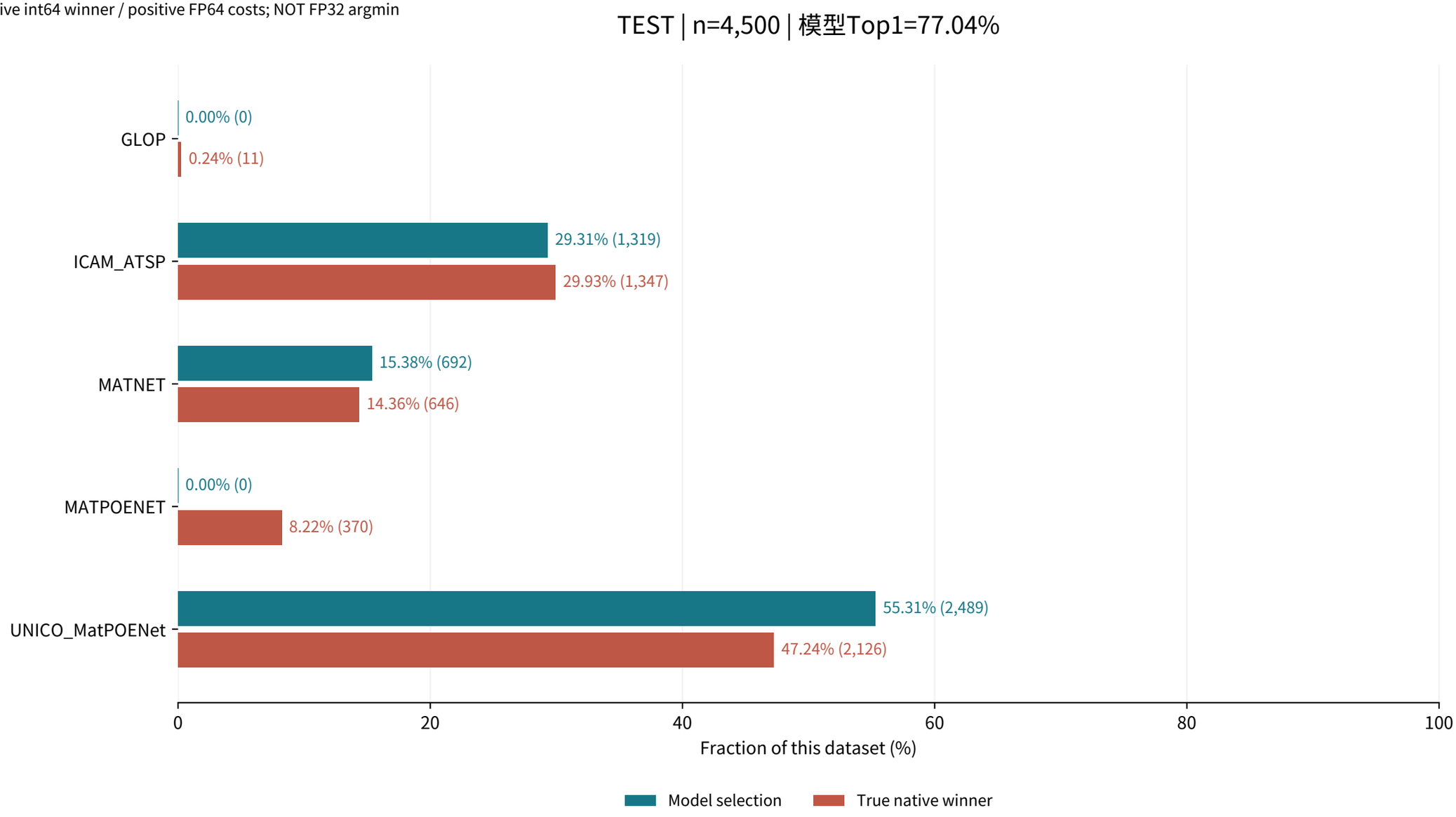
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | ATSP | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

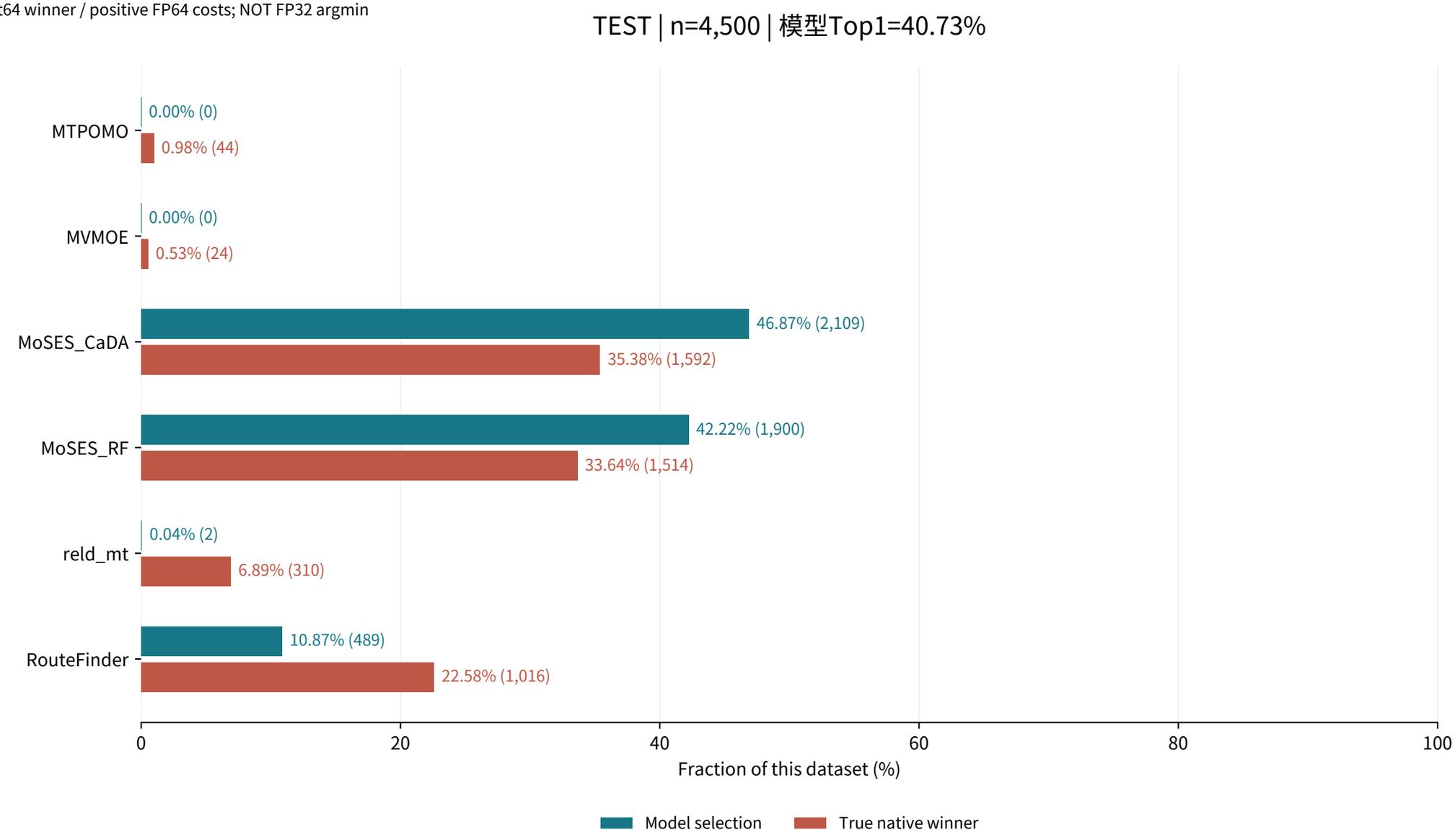
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | OVRP | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

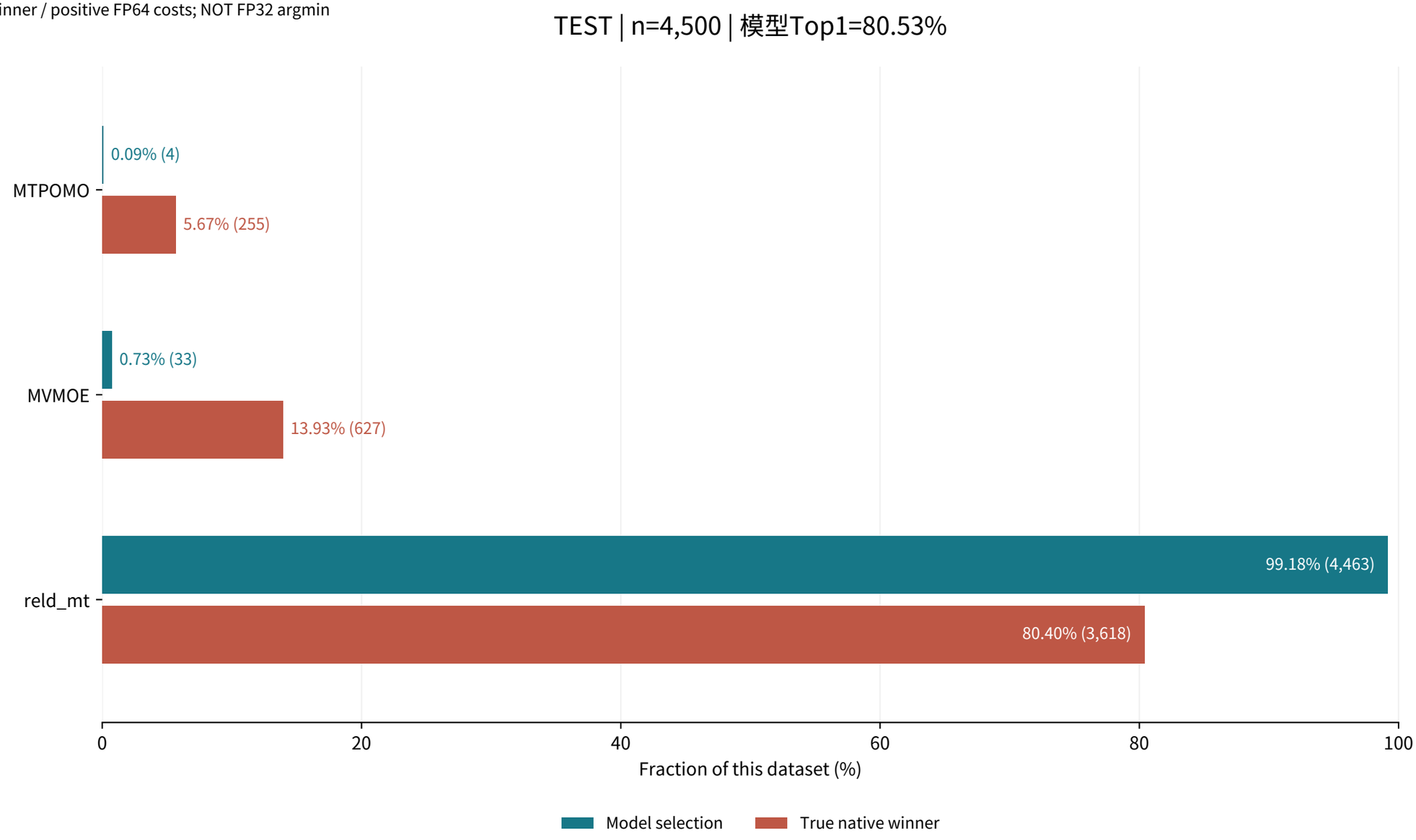
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | VRPB | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

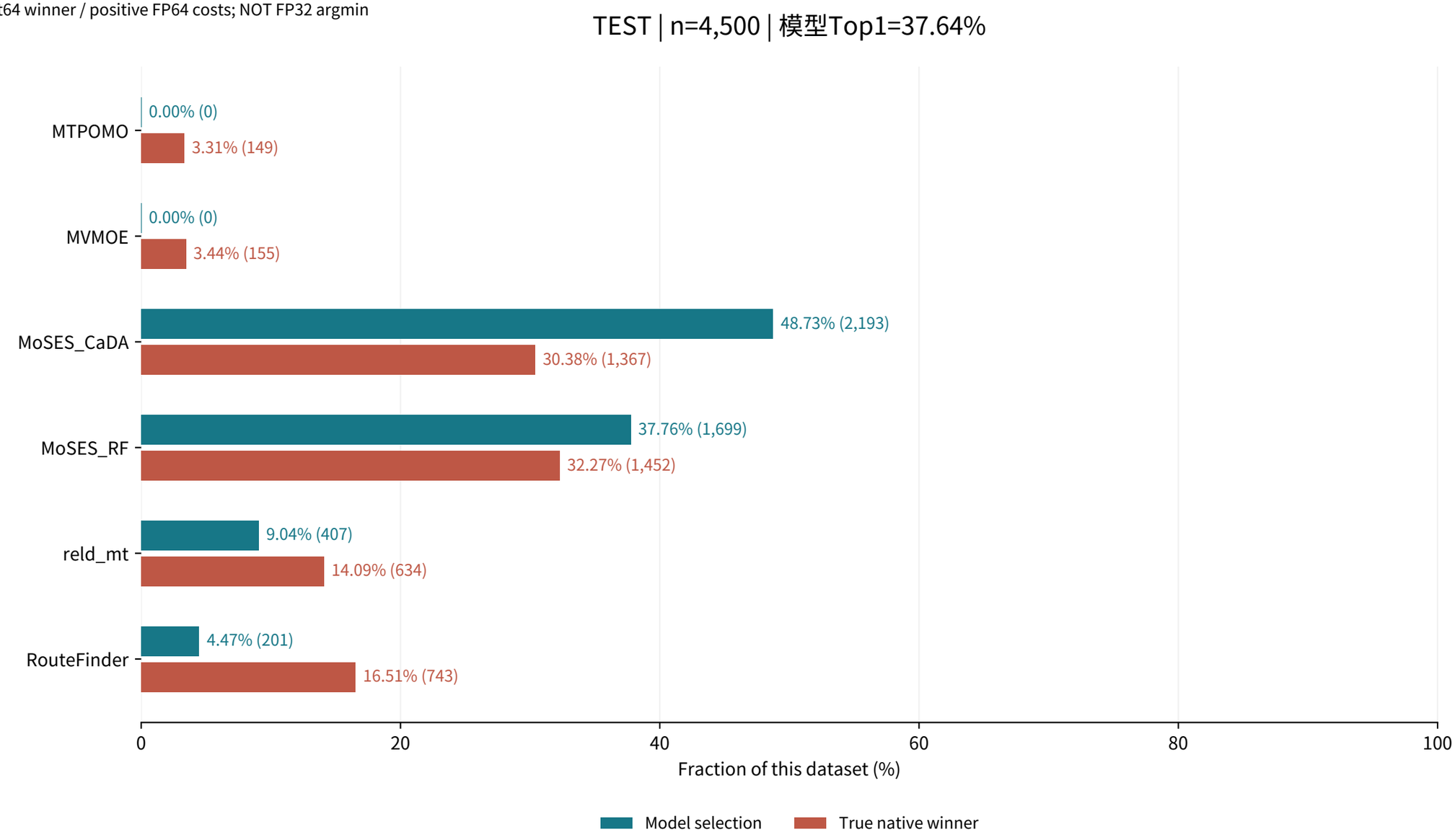
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | VRPL | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

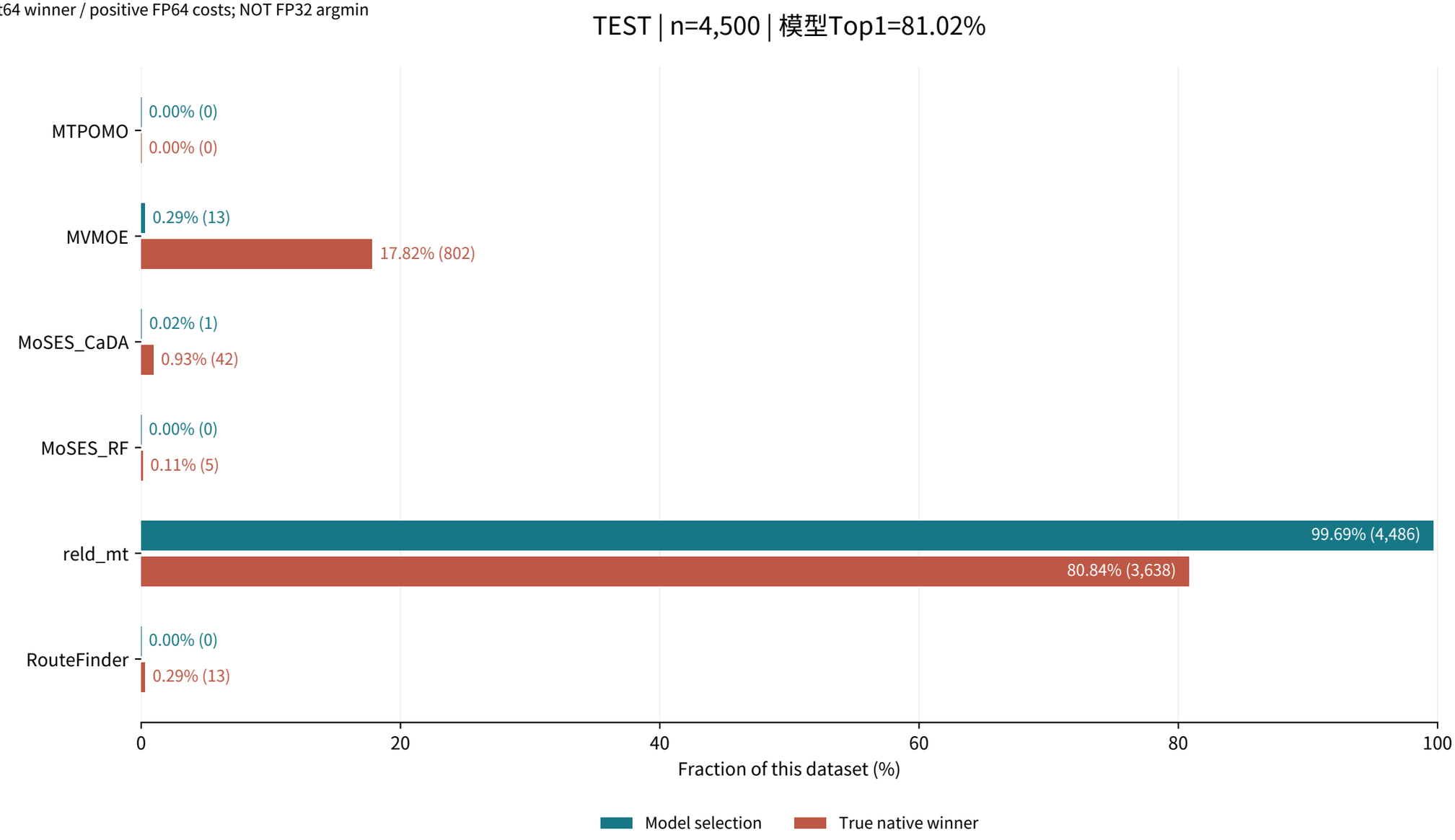
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | VRPTW | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

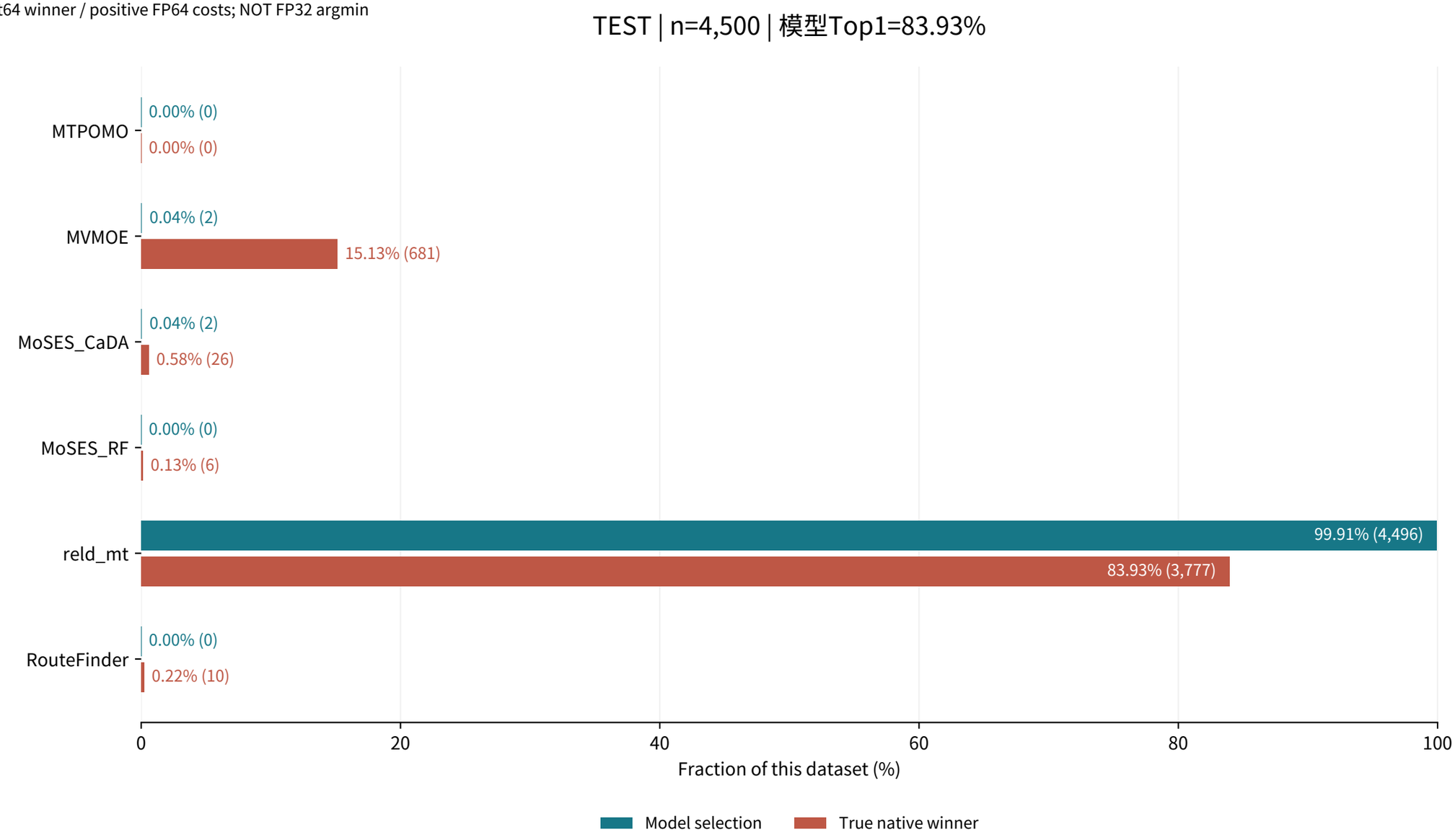
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | OVRPTW | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

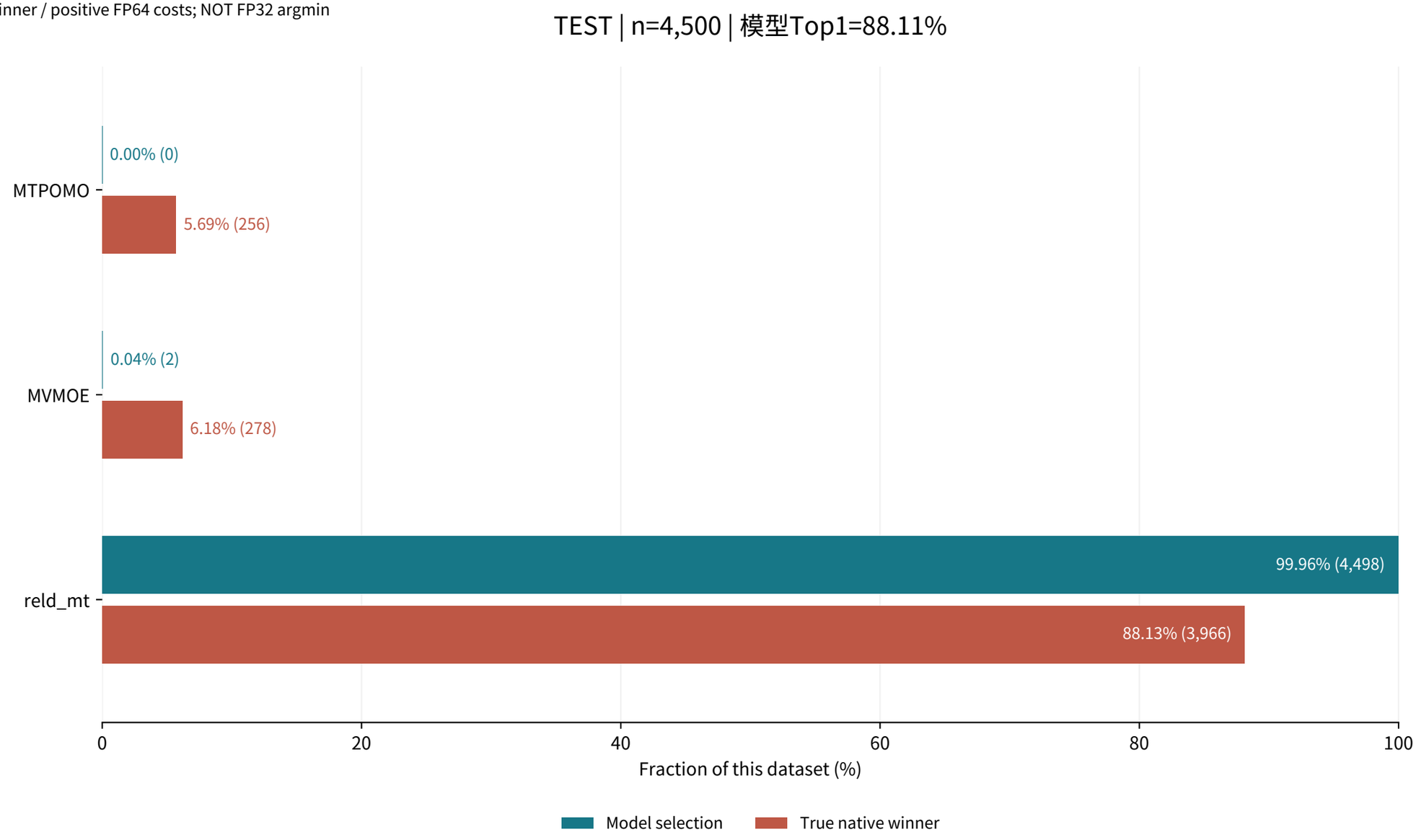
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | OVRPB | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

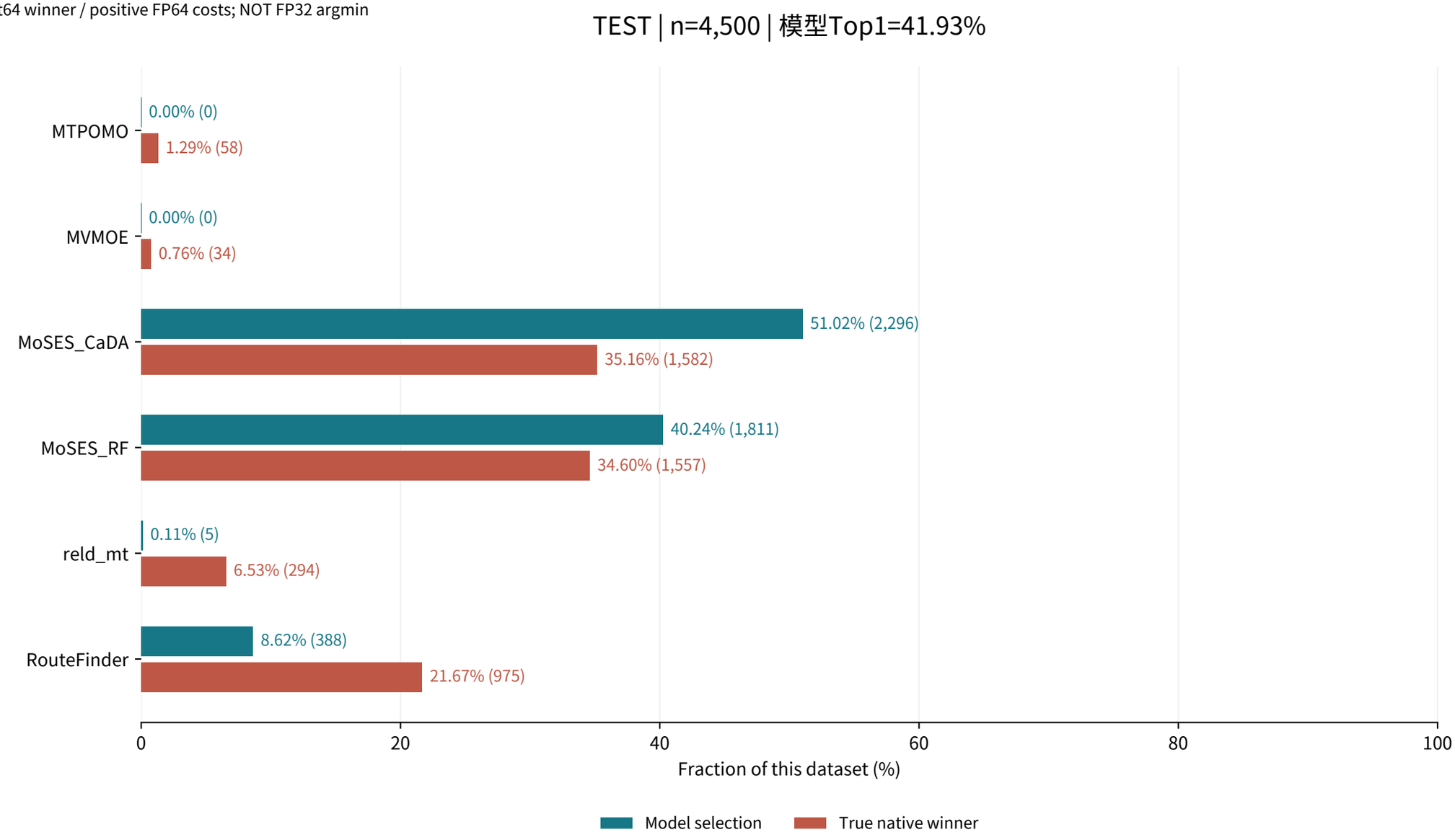
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | OVRPL | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

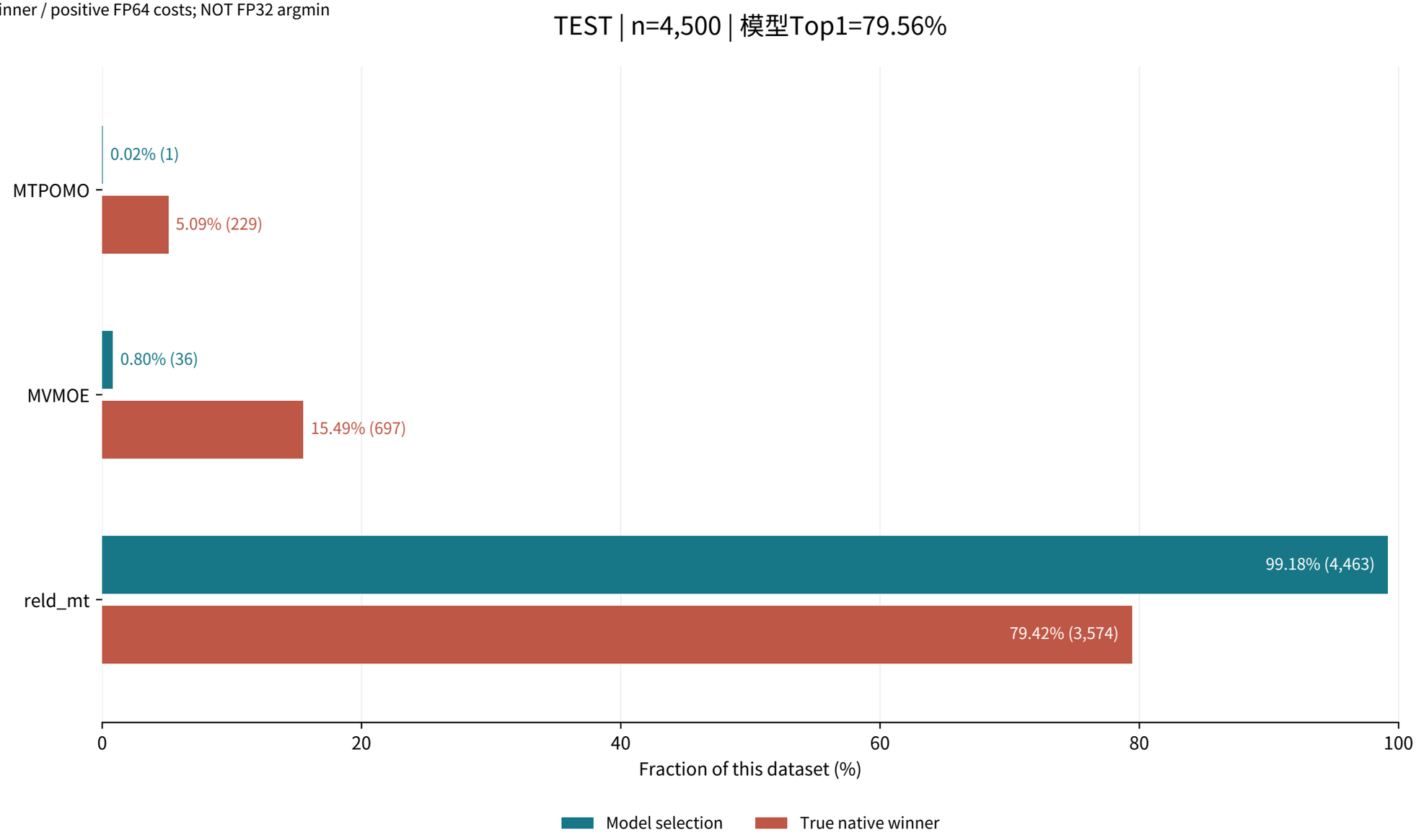
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | VRPBL | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

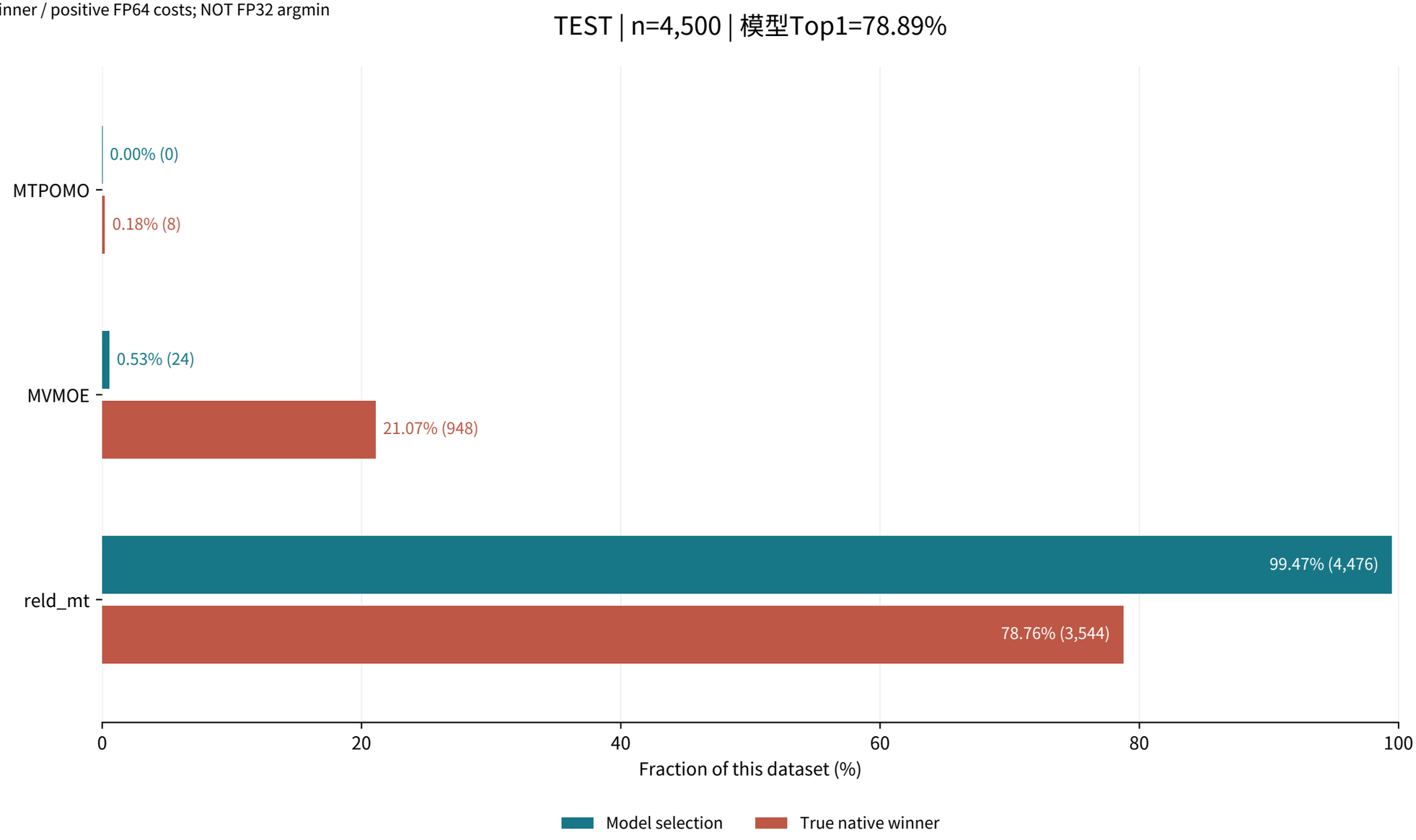
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | VRPBTW | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

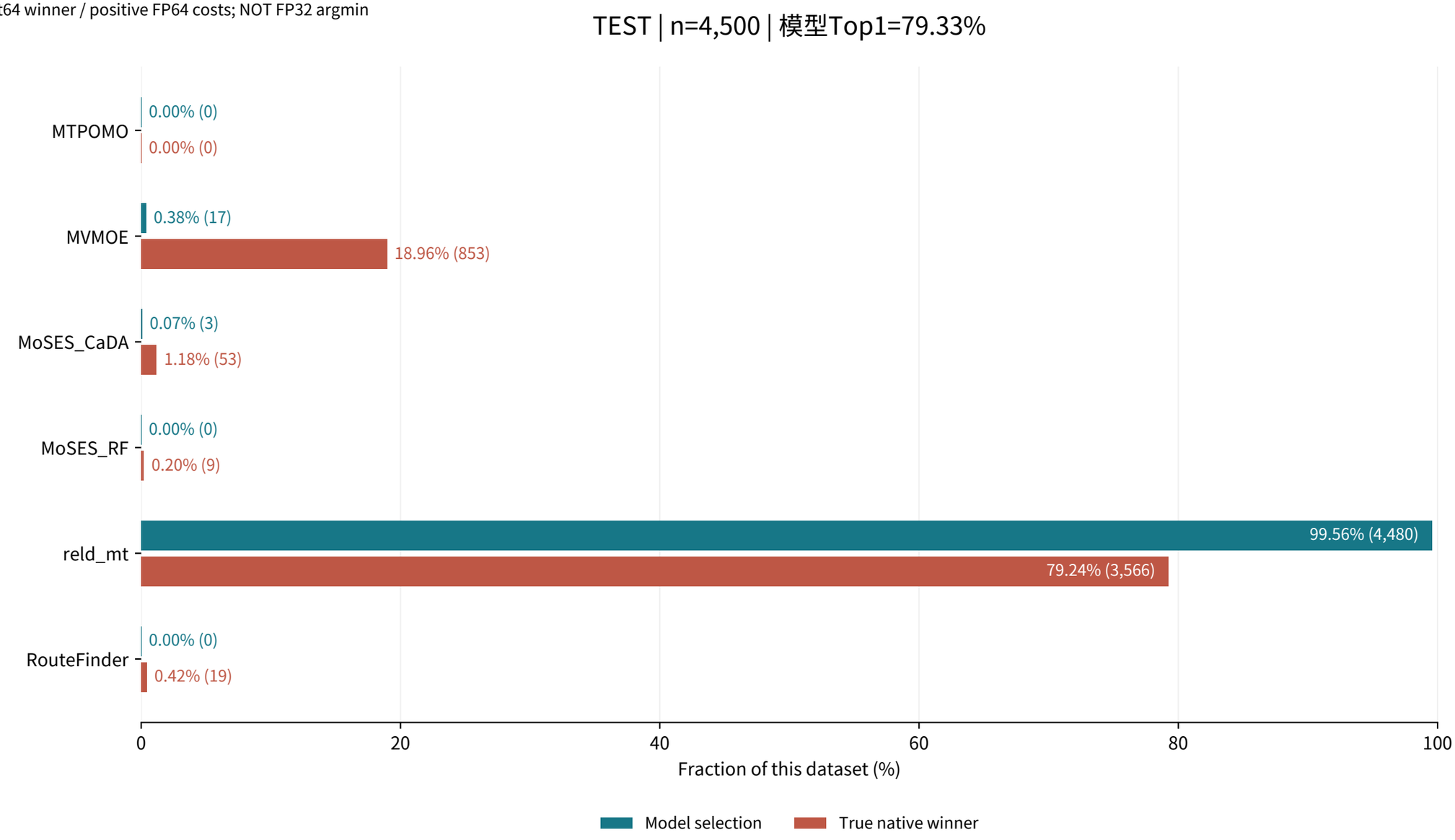
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | VRPLTW | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

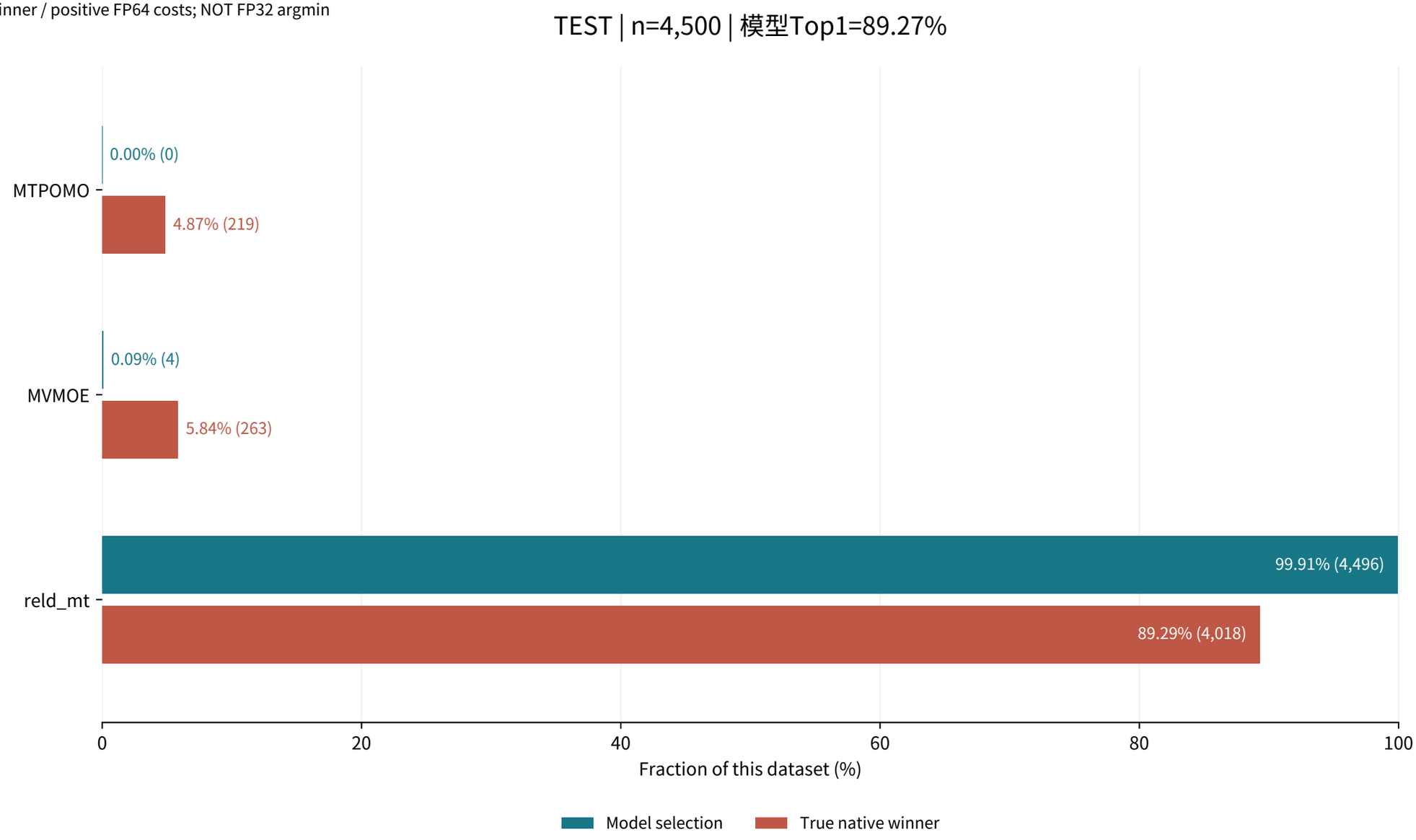
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | OVRPBL | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

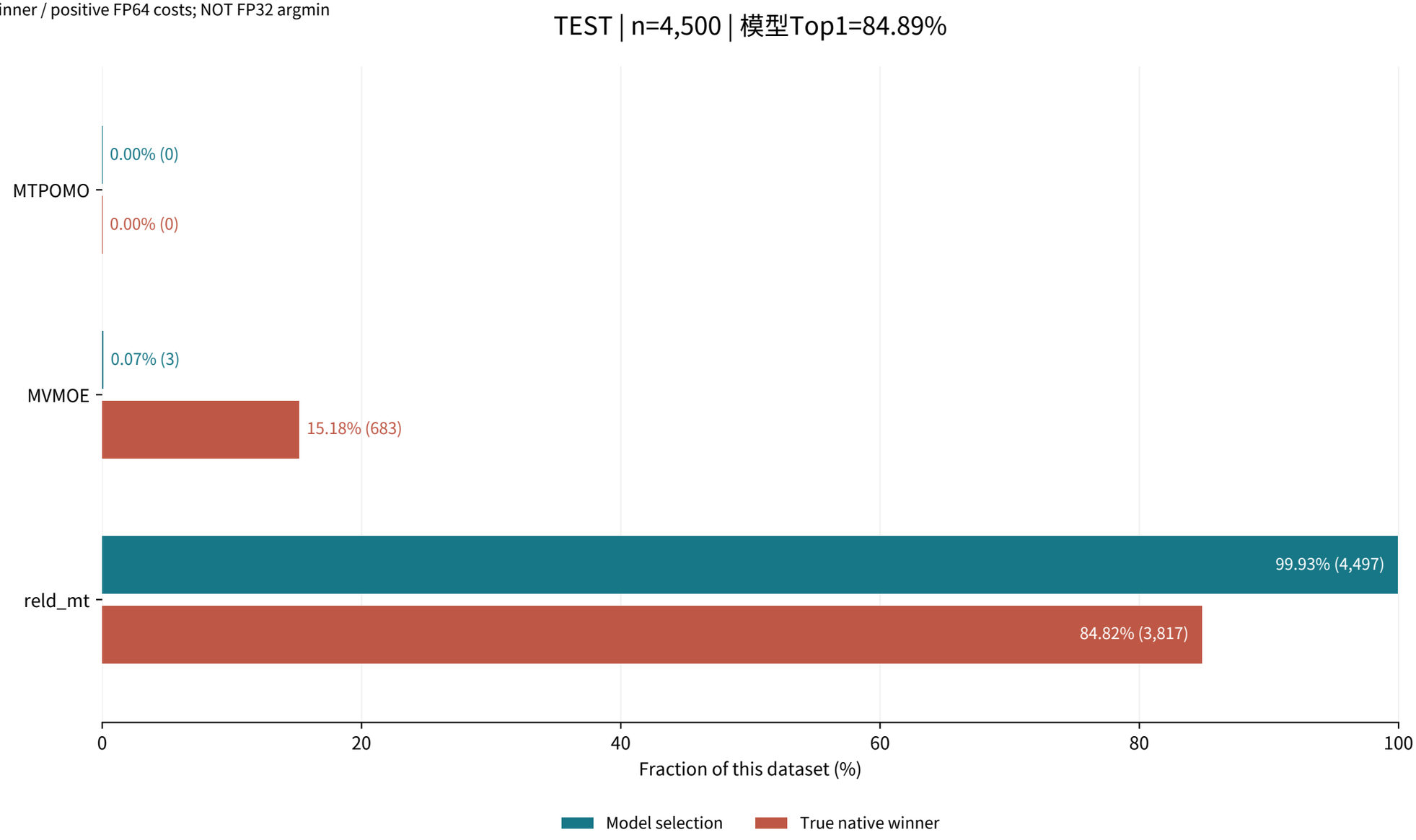
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | OVRPBTW | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

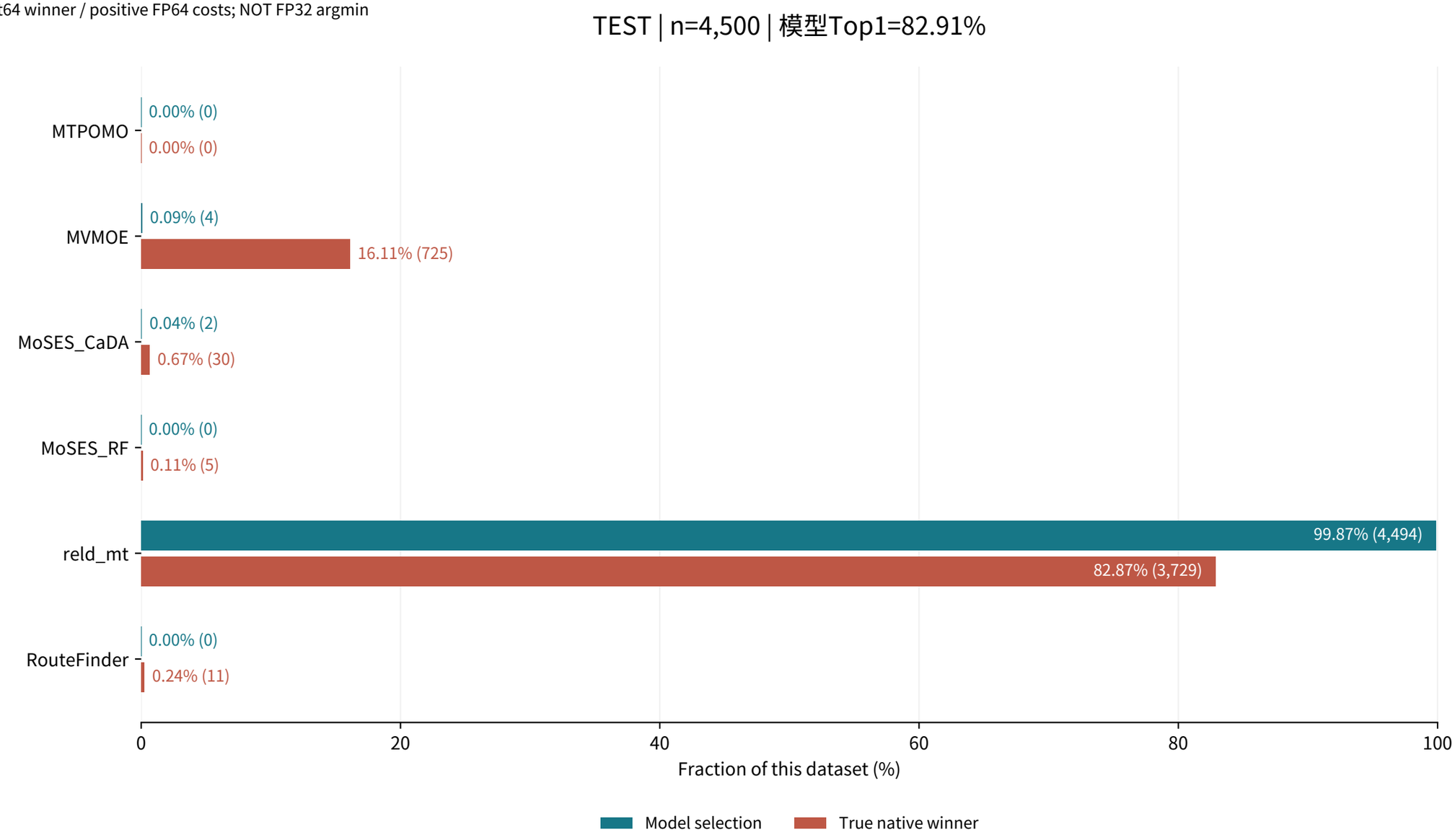
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | OVRPLTW | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

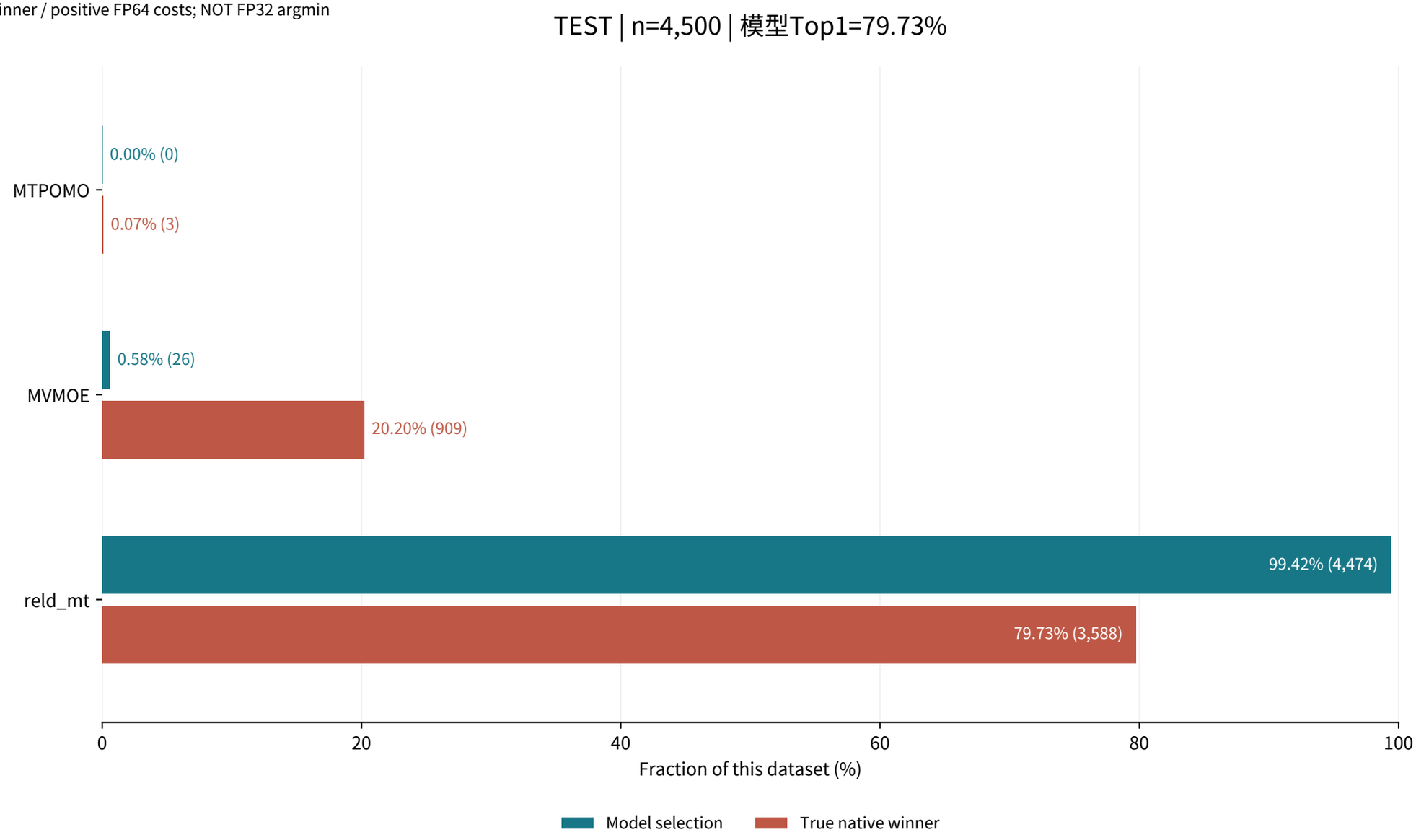
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | VRPBLTW | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

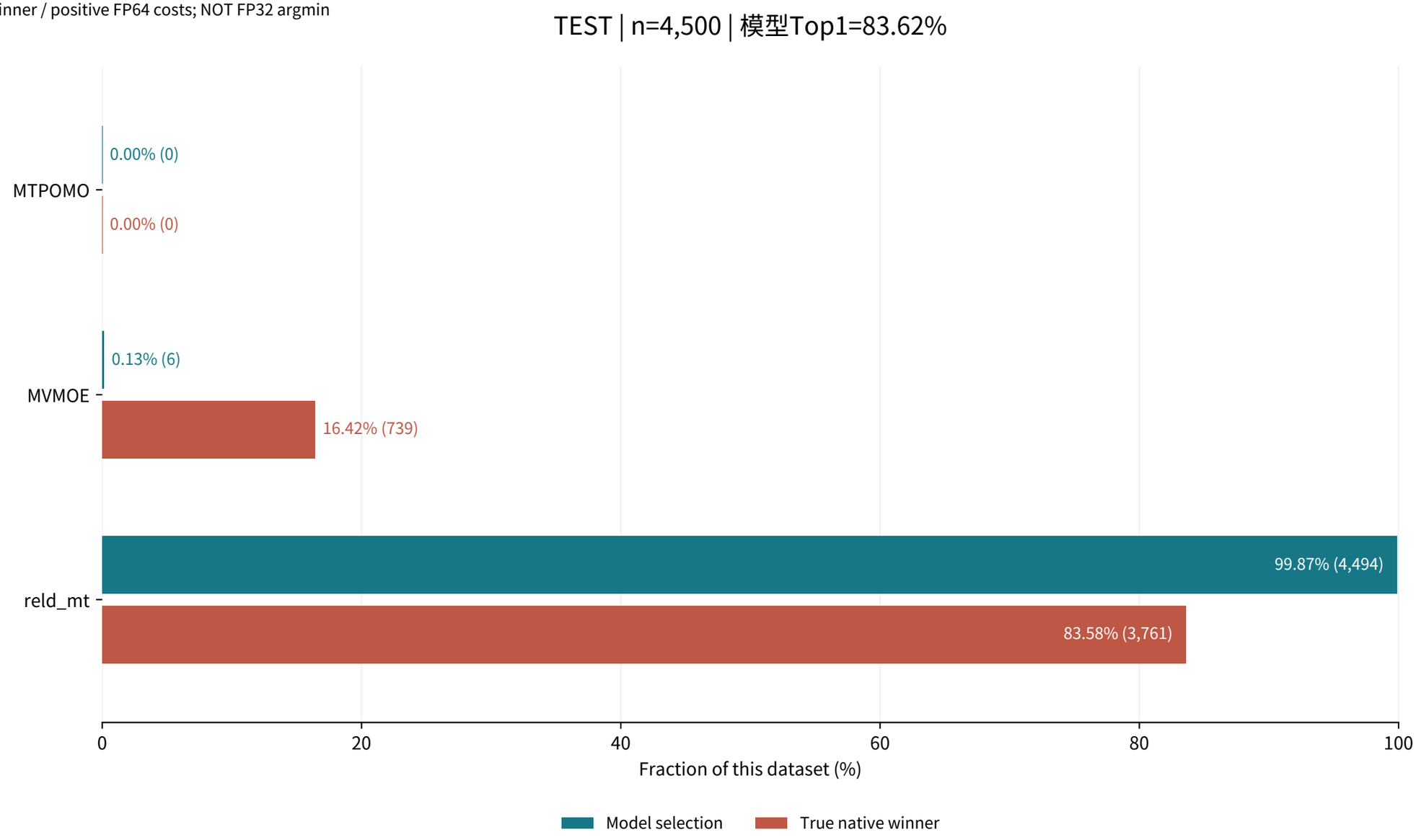
Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 | OVRPBLTW | Model selection vs native winner

Same checkpoint cec914fe375d0b7d60eadacb | selected update 1440

TEST: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018; LIB: original native library

Saved native int64 winner / positive FP64 costs; NOT FP32 argmin



Paired bars use the same axis and fixed candidate order, including zeros. TEST precedes LIB.

Model Top1 is per-instance native-winner agreement, NOT the most frequently selected method.

R123 TEST | Bank coverage vs router errors; cost budget counted once

Task	Bank native cover %	Bank miss pp	Router miss pp	Native error %	Bank floor %	Route excess %	Regret %
TSP	48.733	51.267	2.111	53.378	0.385	0.035	0.421
CVRP	53.222	46.778	5.600	52.378	0.543	0.135	0.678
ATSP	78.400	21.600	1.356	22.956	0.328	0.023	0.351
OVRP	48.533	51.467	7.800	59.267	0.501	0.131	0.633
VRPB	80.867	19.133	0.333	19.467	0.369	0.013	0.382
VRPL	44.000	56.000	6.356	62.356	0.340	0.059	0.399
VRPTW	81.156	18.844	0.133	18.978	0.287	0.004	0.291
OVRPTW	84.000	16.000	0.067	16.067	0.232	0.003	0.236
OVRPB	88.200	11.800	0.089	11.889	0.295	0.009	0.304
OVRPL	50.067	49.933	8.133	58.067	0.499	0.125	0.623
VRPBL	79.844	20.156	0.289	20.444	0.403	0.029	0.432
VRPBTW	79.044	20.956	0.156	21.111	0.431	0.007	0.438
VRPLTW	79.467	20.533	0.133	20.667	0.313	0.004	0.317
OVRPBL	89.378	10.622	0.111	10.733	0.262	0.006	0.268
OVRPBTW	84.956	15.044	0.067	15.111	0.292	0.003	0.294
OVRPLTW	82.978	17.022	0.067	17.089	0.265	0.002	0.267
VRPBLTW	80.067	19.933	0.333	20.267	0.439	0.011	0.450
OVRPBLTW	83.667	16.333	0.044	16.378	0.317	0.006	0.322
ALL18	73.143	26.857	1.843	28.700	0.361	0.034	0.395

Native error = bank miss + covered-but-missed route, with disjoint events.
Regret = (best bank action cost - oracle)/oracle + (chosen cost - best bank action cost)/oracle.
Bank best/coverage are answer-dependent DIAGNOSTICS, not deployable oracle policies.
One locked full18 checkpoint; no TEST/LIB policy or checkpoint selection.
TRAIN here = source-record-certified metaTRAIN10886/task, not full TRAIN81000; historical holdouts development-exposed.

R123 TEST | TRAIN-only fixed-expert and task-conditioned prior references

Task	Router Top1 %	Global fixed %	Task prior %	Router regret %	Global regret %	Prior regret %
TSP	46.62	46.69	46.69	0.421	0.420	0.420
CVRP	47.62	47.93	47.22	0.678	0.687	0.684
ATSP	77.04	77.04	77.04	0.351	0.351	0.351
OVRP	40.73	40.76	40.76	0.633	0.632	0.632
VRPB	80.53	80.62	80.60	0.382	0.383	0.381
VRPL	37.64	37.73	37.18	0.399	0.399	0.405
VRPTW	81.02	81.02	81.02	0.291	0.289	0.289
OVRPTW	83.93	83.96	83.93	0.236	0.253	0.236
OVRPB	88.11	88.13	88.13	0.304	0.303	0.298
OVRPL	41.93	41.22	42.62	0.623	0.631	0.616
VRPBL	79.56	79.53	79.58	0.432	0.432	0.418
VRPBTW	78.89	78.82	78.82	0.438	0.452	0.439
VRPLTW	79.33	79.33	79.33	0.317	0.317	0.317
OVRPBL	89.27	89.27	89.27	0.268	0.269	0.269
OVRPBTW	84.89	84.87	84.89	0.294	0.296	0.296
OVRPLTW	82.91	82.93	82.91	0.267	0.266	0.267
VRPBLTW	79.73	79.67	79.80	0.450	0.466	0.445
OVRPBLTW	83.62	83.53	83.53	0.322	0.327	0.327
ALL18	71.30	71.28	71.30	0.395	0.398	0.394

References fitted only on the SAME certified metaTRAIN bank native-correctness labels.
Global fixed maximizes equal-task TRAIN correctness; task prior maximizes within-task TRAIN correctness.
Task prior is constant within a task, not another instance model; ties use fixed R77B/R80A/R83 order.
One locked full18 checkpoint; no TEST/LIB policy or checkpoint selection.
TRAIN here = source-record-certified metaTRAIN10886/task, not full TRAIN81000; historical holdouts development-exposed.

Same locked R123 vs saved NSS | TEST and native LIB

Dataset	N	R123 Top1 %	NSS Top1 %	R123 cost	NSS cost	NSS status
TSP TEST	4500	46.62	45.89	7.9715	7.9704	complete
CVRP TEST	4500	47.62	45.98	18.011	18.005	complete
TSPLIB	49	18.37	20.41	8.2184	8.2179	complete
CVRPLIB	100	56.00	70.00	67.43	67.101	complete

Same instances, native FP64 labels, named legal pool and checkpoint verified when NSS is available.
NSS is the historical70114 model, NOT a new90k retraining; no new NSS inference.
One locked full18 checkpoint; no TEST/LIB policy or checkpoint selection.
TRAIN here = source-record-certified metaTRAIN10886/task, not full TRAIN81000; historical holdouts development-exposed.

Actual live hard routing | Gate plus ONE selected complete expert per query

Dataset	N	Batch	FP32 forward sec	ms / query	Executed expert rows
TSPLIB	49	UNKNOWN	UNKNOWN	UNKNOWN	[0, 42, 7]
CVRPLIB	100	1	0.9861	9.8607	[0, 94, 6]
TSP TEST prefix	128	128	0.0999	0.7803	[0, 112, 16]
CVRP TEST prefix	128	128	0.1178	0.9206	[0, 99, 29]
ATSP TEST prefix	128	128	0.0155	0.1209	[0, 128, 0]
OVRP TEST prefix	128	128	0.0208	0.1624	[0, 80, 48]
VRPB TEST prefix	128	128	0.0209	0.1631	[0, 100, 28]
VRPL TEST prefix	128	128	0.0208	0.1623	[0, 104, 24]
VRPTW TEST prefix	128	128	0.0210	0.1641	[0, 96, 32]
OVRPTW TEST prefix	128	128	0.0214	0.1672	[0, 105, 23]
OVRPB TEST prefix	128	128	0.0213	0.1660	[0, 124, 4]
OVRPL TEST prefix	128	128	0.0212	0.1660	[0, 104, 24]
VRPBL TEST prefix	128	128	0.0218	0.1707	[0, 126, 2]
VRPBTW TEST prefix	128	128	0.0210	0.1642	[0, 89, 39]
VRPLTW TEST prefix	128	128	0.0219	0.1712	[0, 114, 14]
OVRPBL TEST prefix	128	128	0.0219	0.1708	[0, 125, 3]
OVRPBTW TEST prefix	128	128	0.0212	0.1653	[0, 95, 33]
OVRPLTW TEST prefix	128	128	0.0214	0.1669	[0, 103, 25]
VRPBLTW TEST prefix	128	128	0.0213	0.1664	[0, 107, 21]
OVRPBLTW TEST prefix	128	128	0.0212	0.1658	[0, 112, 16]

Direct live router calls have NO expert-score input. Dispatch hooks require total expert rows = query count.
Forward includes gate, chosen expert and dispatch. One warmup batch excluded; no parameter update.
Geometry/input preparation, gather/transfer/output copy and three-bank cache construction excluded and recorded separately.
Natural timing uses fixed first TEST rows/task only, not a performance-selected sample or tuning.
UNKNOWN: a legacy capture survived a failed provenance write but elapsed timing did not; capture reused, not remeasured.

Six descriptive zero / near-zero lists | Native winner support retained

Split	Group	Methods	Native wins	Missed native wins	ALL18 regret contribution pp
train	zero_selected_with_winner	25	3177	3177	0.0196
train	near_zero_selected_with_winner	12	6758	6730	0.0717
val	zero_selected_with_winner	28	2222	2222	0.0438
val	near_zero_selected_with_winner	8	2348	2337	0.0511
test	zero_selected_with_winner	24	1880	1880	0.0355
test	near_zero_selected_with_winner	13	4337	4321	0.0927

Strict zero: no picks and positive winner support. Near zero: (0, 0.1%] picks and positive winner support.
Lists are disjoint within each split; full methods.csv retains ALL candidates, including zero-winner candidates.
Regret is charged once to the missed native-winner row. No false-pick double charge; no frequency quotas.
One locked full18 checkpoint; no TEST/LIB policy or checkpoint selection.
TRAIN here = source-record-certified metaTRAIN10886/task, not full TRAIN81000; historical holdouts development-exposed.